```
package com.polarbookshop.catalogservice.persistence;

import java.util.Map;
import java.util.Optional;
import java.util.concurrent.ConcurrentHashMap;
import com.polarbookshop.catalogservice.domain.Book;
import com.polarbookshop.catalogservice.domain.BookRepository;
import org.springframework.stereotype.Repository;

@Repository                                                          ❶
public class InMemoryBookRepository implements BookRepository {
  private static final Map<String, Book> books =                     ❷
    new ConcurrentHashMap<>();

  @Override
  public Iterable<Book> findAll() {
    return books.values();
  }

  @Override
  public Optional<Book> findByIsbn(String isbn) {
    return existsByIsbn(isbn) ? Optional.of(books.get(isbn)) :
      Optional.empty();
  }

  @Override
  public boolean existsByIsbn(String isbn) {
    return books.get(isbn) != null;
  }

  @Override
  public Book save(Book book) {
    books.put(book.isbn(), book);
    return book;
  }

  @Override
  public void deleteByIsbn(String isbn) {
    books.remove(isbn);
  }
}
```

❶ Stereotype annotation that marks a class to be a repository managed by Spring

❷ In-memory map to store books for testing purposes

## *USING EXCEPTIONS TO SIGNAL ERRORS IN THE DOMAIN*